Show (well), Attend (better), and Tell (hopefully correctly)

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Introduction

Image-captioning requires human-like models that can detect many objects in images and compress salient information into concise, semantically-meaningful captions.

Motivation

- Prior work: *Show and Tell* used a CNN to encode an image into a single vector, then generated a caption with an RNN.
- Limitation: Loses spatial info that could be useful for richer captions.
- *Show, Attend, Tell* introduced **attention** to focus on salient regions of an image during caption generation, improving performance and interpretability.

Goals

- Train and validate caption generators using **soft** and **hard** attention mechanisms on the **Flickr8k** Image-Caption dataset.
- Evaluate the impact of **feature representation** on caption quality.

Human-generated caption: Professor Weinberger hits a yoda piñata.

Paper Summary

Show, Attend and Tell proposes an attention-based deep neural network model for image caption generation. The model combines a CNN encoder for feature extraction with an LSTM decoder which produces the output caption step-by-step. It introduces an attention mechanism with two variants: 1) soft and 2) hard. The model is evaluated on three datasets of images and captions: Flickr8k, Flickr30k and MS COCO.

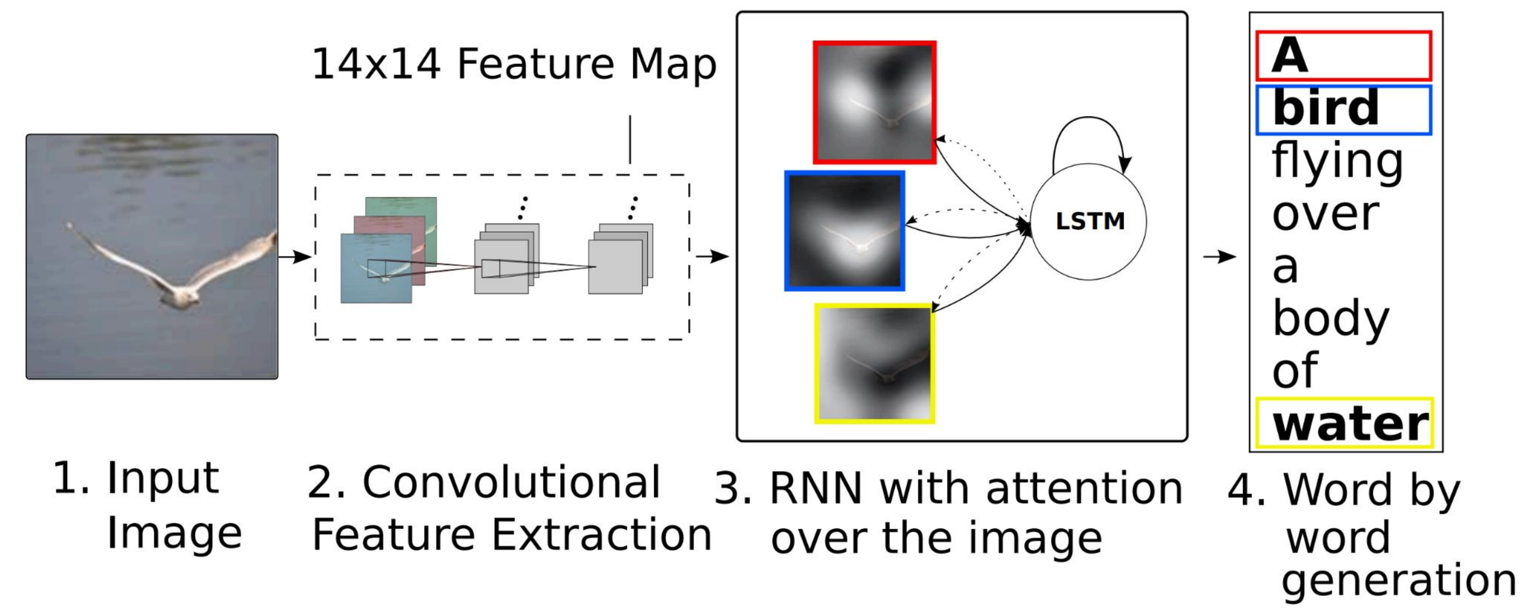
Methods

Flickr8k Dataset:

- 8K images
- 5 captions per image

$$e_{ti} = f_{\text{att}}(\mathbf{a}_i, \mathbf{h}_{t-1})$$

$$\alpha_{ti} = \frac{\exp(e_{ti})}{\sum_{k=1}^{L} \exp(e_{tk})}$$



Soft Attention

$$\mathbb{E}_{p(s_t|a)}[\hat{\mathbf{z}}_t] = \sum_{i=1}^{L} \alpha_{t,i}\mathbf{a}_i$$

Doubly Stochastic Attention

$$L_d = -\log(P(\mathbf{y}|\mathbf{x})) + \lambda \sum_{i}^{L}(1 - \sum_{t}^{C} \alpha_{ti})^2$$

Hard Attention

$$\tilde{s}_t \sim \text{Multinoulli}_L(\{\alpha_i\})$$

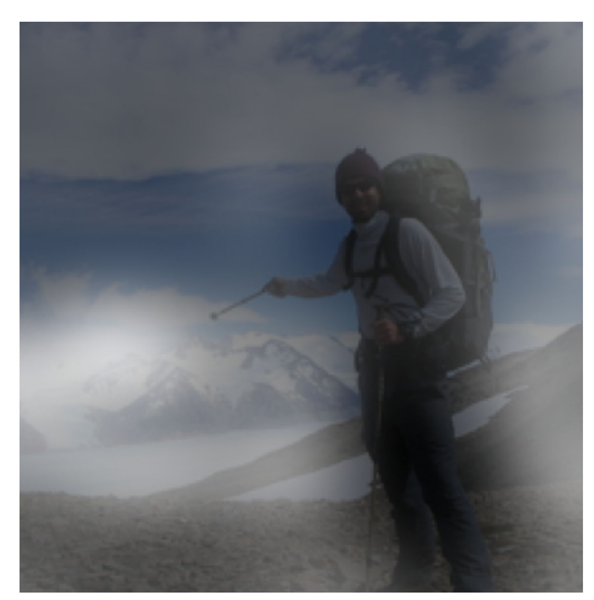
$$p(s_{t,i} = 1 \mid s_{j<t}, \mathbf{a}) = \alpha_{t,i}$$

$$\hat{\mathbf{z}}_t = \sum_i s_{t,i}\mathbf{a}_i.$$

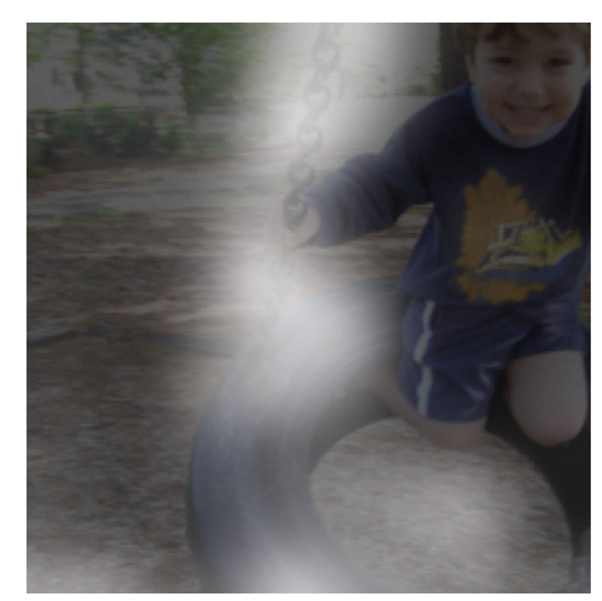
Trained using REINFORCE

What does the model "see"?

A dog is playing in the grass. ✓



A man in a blue shirt is standing on a rock overlooking a snowy mountain. ✓



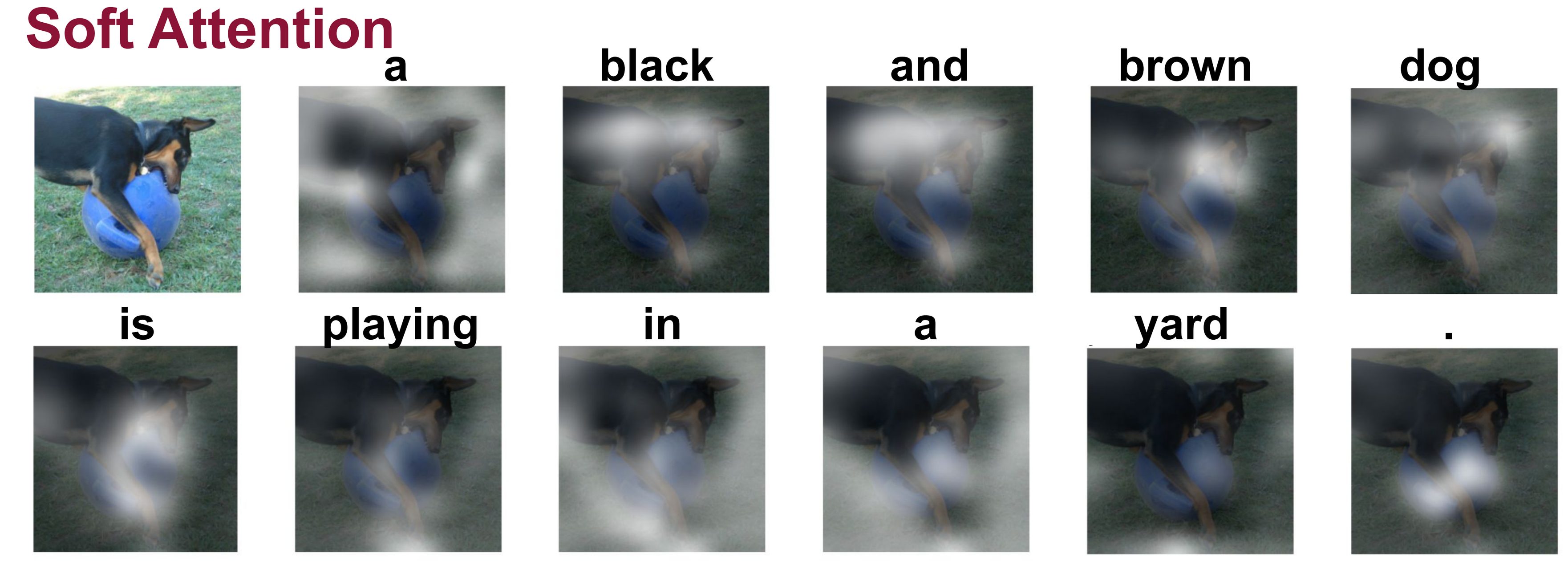
A boy in a blue shirt is riding a bike on a tire swing. ✗

Results

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR
Soft-Attention VGG (Xu, K., et al., 2015)	67.00	44.80	29.90	19.50	18.93
Soft-Attention VGG	65.67	45.67	32.14	22.61	20.72
Soft-Attention ResNet	**68.01**	**48.30**	**34.39**	**24.20**	**21.66**
Hard-Attention VGG (Xu, K., et al., 2015)	67.00	45.70	31.40	21.30	20.30
Hard-Attention VGG	64.72	43.57	29.87	20.46	19.28
Hard-Attention ResNet	**68.31**	**48.75**	**34.67**	**24.40**	**22.04**

Table 1. BLEU-1,2,3,4/METEOR metrics on Flickr8k Dataset.

Soft Attention



Hard Attention

Conclusion

- Attention improves quality/interpretability and we outperformed paper results.
- **Stronger feature extractors (ResNet50) produce better captions than VGG19.**
- Hard attention is harder to train than soft attention due to its stochastic nature.

References

Xu, K., Ba, J., Kiros, R., Cho, K., Courville, A., Salakhudinov, R., ... & Bengio, Y. (2015, June). Show, attend and tell: Neural image caption generation with visual attention. In *International conference on machine learning* (pp. 2048-2057). PMLR.

Hodosh, M., Young, P., & Hockenmaier, J. (2013). Framing image description as a ranking task: Data, models and evaluation metrics. *Journal of Artificial Intelligence Research*, 47, 853–899.

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